

Chromatic thresholds for linear equations

Roth meets Erdős–Simonovits via Cayley graphs

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Linear patterns in finite fields

Fix a homogeneous linear equation

$$\mathcal{L}: \quad c_1 x_1 + \cdots + c_k x_k = 0, \quad k \geq 3,$$

with nonzero integer coefficients.

We work in $\mathbb{F}_p$, and call $A \subseteq \mathbb{F}_p$ **$\mathcal{L}$ -solution-free** if there is no solution

$$x_1, \dots, x_k \in A$$

with pairwise distinct coordinates.

Basic extremal question

How large can an $\mathcal{L}$ -solution-free set be?

Classical examples

3-term AP

$$x_1 - 2x_2 + x_3 = 0.$$

Every dense set has a solution.

Schur equation

$$x_1 + x_2 - x_3 = 0.$$

Exists a solution-free set with density $1/3$.

The coefficient sums already predict very different density behaviour.

Roth's Theorem

Theorem (Roth)

For fixed $\mathcal{L}$, the following are equivalent:

- 1 every $\mathcal{L}$ -solution-free set $A \subseteq \mathbb{F}_p$ has size $o(p)$;
- 2 $c_1 + \dots + c_k = 0$.

$\sum_i c_i \neq 0$
interval obstruction



positive-density
solution-free sets exist

$\sum_i c_i = 0$
Roth theorem



every positive-density set
contains a solution

What Roth does not say

If $c_1 + \cdots + c_k \neq 0$, a dense solution-free set A may exist.

Roth's theorem does not tell us about the structure of A .

Refined question

Suppose $A \subseteq \mathbb{F}_p$ is dense and $\mathcal{L}$ -solution-free.

Does A still have to possess some global structure?

Our structure will be measured by the Cayley graph generated by A .

Cayley graphs generated by additive sets

For $A \subseteq \mathbb{F}_p$, define

$G_A :=$ the underlying graph of $\text{Cay}(\mathbb{F}_p, A)$.

The vertices are the elements of $\mathbb{F}_p$.

Two distinct vertices u, v are adjacent if

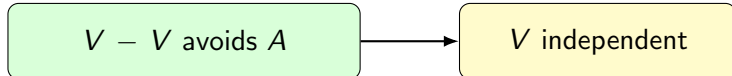
$$v - u \in A \quad \text{or} \quad u - v \in A.$$

Independent sets

A set $V \subseteq \mathbb{F}_p$ is independent in G_A iff

$$((V - V) \setminus \{0\}) \cap A = \emptyset.$$

So an independent set is a set whose internal differences avoid A .



Additive Ramsey–Turan Result

Theorem (Bucić–Christoph–Kim–Lee–Sivashankar)

For fixed $\mathcal{L} : \sum_{i=1}^k c_i x_i = 0$, the following are equivalent:

every $\mathcal{L}$ -solution-free $A \subseteq \mathbb{F}_p$ with $|A| \geq \varepsilon p$ has $\alpha(\text{Cay}(\mathbb{F}_p, A)) \geq cp$

for some $c = c(\varepsilon, \mathcal{L}) > 0$, and

$$\exists \emptyset \neq I \subseteq [k] \text{ such that } \sum_{i \in I} c_i = 0.$$

This determines when dense solution-free sets forces linear independent sets.

We consider when dense solution-free sets forces bounded chromatic number.

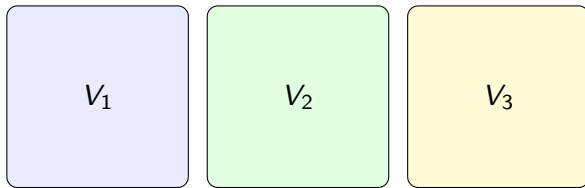
Coloring

A proper coloring of G_A with C colors is a partition

$$\mathbb{F}_p = V_1 \sqcup \cdots \sqcup V_C$$

such that

$$((V_i - V_j) \setminus \{0\}) \cap A = \emptyset \quad \text{for every } i, j.$$



each cell is difference-avoiding

Graph-theoretic chromatic thresholds

For a fixed graph H , the chromatic threshold asks:

$$\delta_\chi(H) = \inf \left\{ d > 0 : \text{every } H\text{-free } n\text{-vertex graph } G \text{ with } \delta(G) \geq dn \text{ has bounded } \chi(G) \right\}.$$

This is a classical Erdős–Simonovits problem.

The additive analogue

We replace graph-theoretic objects by additive ones:

$$\begin{array}{lll} H\text{-free graph} & \longleftrightarrow & \mathcal{L}\text{-solution-free generator } A, \\ \text{minimum degree} & \longleftrightarrow & |A|/p, \\ \chi(G) & \longleftrightarrow & \chi(\text{Cay}(\mathbb{F}_p, A)). \end{array}$$

So the question becomes:

How dense must a solution-free generator be before its Cayley graph has bounded chromatic number?

Definition

Definition

The **chromatic threshold** of $\mathcal{L}$ is

$$\delta_{\chi}(\mathcal{L}) = \inf \left\{ d > 0 : \begin{array}{l} \exists C = C(d, \mathcal{L}) \text{ such that for every prime } p, \\ A \subseteq \mathbb{F}_p, A \text{ } \mathcal{L}\text{-solution-free, } |A| \geq dp \\ \implies \chi(\text{Cay}(\mathbb{F}_p, A)) \leq C \end{array} \right\}.$$

Today we determine exactly when

$$\delta_{\chi}(\mathcal{L}) = 0.$$

Interpreting $\delta_\chi(\mathcal{L}) = 0$

The condition $\delta_\chi(\mathcal{L}) = 0$ means:

for every fixed density $\varepsilon > 0$, there exists $C = C(\varepsilon, \mathcal{L})$ such that

$$|A| \geq \varepsilon p, \quad A \text{ is } \mathcal{L}\text{-solution-free} \implies \chi(\text{Cay}(\mathbb{F}_p, A)) \leq C.$$

Equivalently, every dense solution-free generator admits a bounded partition

$$\mathbb{F}_p = V_1 \sqcup \cdots \sqcup V_C$$

with

$$((V_i - V_i) \setminus \{0\}) \cap A = \emptyset.$$

Main theorem

Theorem

Let

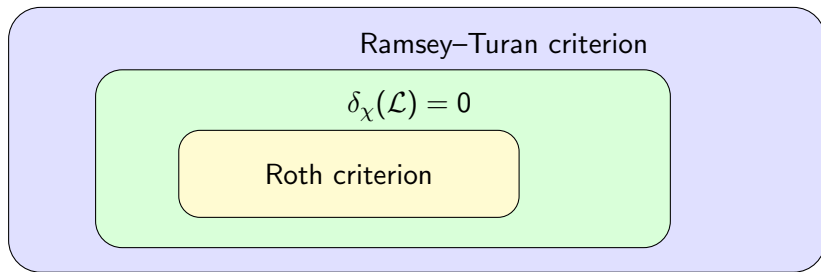
$$\mathcal{L} : \quad c_1 x_1 + \cdots + c_k x_k = 0, \quad k \geq 3,$$

with all $c_i \neq 0$. Then

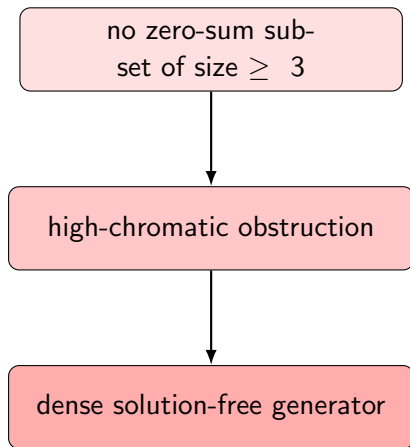
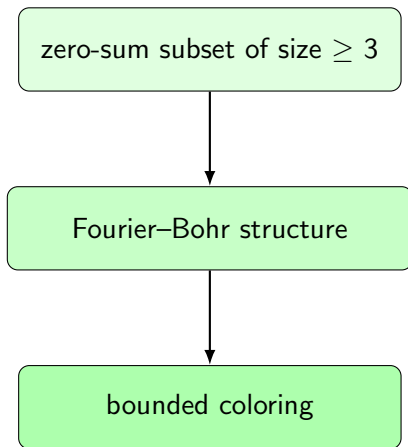
$$\delta_{\chi}(\mathcal{L}) = 0 \quad \Longleftrightarrow \quad \exists I \subseteq [k], \quad |I| \geq 3, \quad \sum_{i \in I} c_i = 0.$$

The three regimes

notion	coefficient condition	structural consequence
Roth density	$\sum_{i \in [k]} c_i = 0$	no dense solution-free sets
chromatic threshold 0	$\exists I, I \geq 3, \sum_{i \in I} c_i = 0$	bounded global coloring
Ramsey–Turan type	$\exists I \neq \emptyset, \sum_{i \in I} c_i = 0$	one large independent set



Proof architecture



Positive direction: statement

Assume there is a zero-sum subcollection

$$I \subseteq [k], \quad |I| \geq 3, \quad \sum_{i \in I} c_i = 0.$$

Let $A \subseteq \mathbb{F}_p$ be dense and $\mathcal{L}$ -solution-free:

$$|A| \geq \varepsilon p.$$

We want to show

$$\chi(\text{Cay}(\mathbb{F}_p, A)) \leq C(\varepsilon, \mathcal{L}).$$

The proof finds a bounded-rank Bohr set B such that A has only boundedly many points in B .

A Fourier-analytic language

For a function $f : \mathbb{F}_p \rightarrow \mathbb{C}$, define

$$\widehat{f}(\xi) = \frac{1}{p} \sum_{x \in \mathbb{F}_p} f(x) e_p(-\xi x), \quad e_p(t) = e^{2\pi i t/p}.$$

Two basic identities:

$$f(x) = \sum_{\xi \in \mathbb{F}_p} \widehat{f}(\xi) e_p(\xi x), \quad \sum_{\xi \in \mathbb{F}_p} |\widehat{f}(\xi)|^2 = \frac{1}{p} \sum_{x \in \mathbb{F}_p} |f(x)|^2.$$

For $A \subseteq \mathbb{F}_p$, the large Fourier coefficients

$$\text{Spec}_\nu(A) = \{\xi : |\widehat{1_A}(\xi)| \geq \nu\}$$

record the structured directions of A .

In the proof: $\text{Spec}_\nu(A) \rightsquigarrow$ a bounded-rank Bohr set $\rightsquigarrow$ a bounded coloring.

Large spectrum

Define the large spectrum of A :

$$L = \{\xi \in \mathbb{F}_p : |\widehat{1_A}(\xi)| \geq \nu\}.$$

Parseval gives

$$|L| \leq \nu^{-2}.$$

Thus the large spectrum has bounded size depending only on ε and $\mathcal{L}$.

This bounded set of characters will define a bounded-rank Bohr set.

Let $\| \cdot \|_{\mathbb{T}}$ denote the distance to the nearest integer.

For a bounded set of characters Γ , define

$$B(\Gamma, \rho) = \{x \in \mathbb{F}_p : \|\xi x/p\|_{\mathbb{T}} \leq \rho \text{ for all } \xi \in \Gamma\}.$$

Intuition:

- elements of B are approximate periods for the structured part of 1_A ;
- if y is built from elements of B , then $e_p(y\xi)$ is close to 1 on the large spectrum.

In the proof, Γ is a coefficient-rescaled version of L .

The key claim

Key claim

Let p be sufficiently large. For the Bohr set $B := B(\Gamma, \rho)$, where $\Gamma = c_s^{-1}L$ for some $s \in I$, one has

$$|A \cap B| < k$$

This is the structural heart of the positive direction.

It says: a dense solution-free generator cannot cluster inside the Bohr set controlled by its own large Fourier spectrum.

The key claim

Suppose $A \cap B$ contains enough points to choose the variables outside I :

$$x_j \in A \cap B, \quad j \in [k] \setminus I.$$

Set

$$y = - \sum_{j \notin I} c_j x_j.$$

Compare the number of solutions in A' to

$$\sum_{i \in I} c_i x_i = 0 \quad \text{and} \quad \sum_{i \in I} c_i x_i = y.$$

Fourier analysis shows these two counts are close, because y is almost invisible on the large spectrum and the small spectrum is controlled by Parseval.

Completing the contradiction

The zero right-hand side equation has many solutions by supersaturation.

The Fourier comparison implies that the shifted equation also has many solutions:

$$\sum_{i \in I} c_i x_i = y.$$

After discarding tuples with repeated coordinates, we find distinct elements of A satisfying

$$\sum_{i \in I} c_i x_i = - \sum_{j \notin I} c_j x_j.$$

That is a forbidden $\mathcal{L}$ -solution in A .

Hence $|A \cap B|$ must be bounded.

Where $|I| \geq 3$ is used

Outside the large spectrum, one Fourier factor is small:

$$|\widehat{1_A}(c_s \xi)| \leq \nu.$$

We need to control

$$\sum_{\xi \in \mathbb{F}_p} \prod_{i \in I \setminus \{s\}} |\widehat{1_A}(c_i \xi)|.$$

If $|I| \geq 3$, at least two factors remain, and Hölder plus Parseval gives the required bound.

This is exactly where a cancelling pair is insufficient.

From Bohr structure to coloring

Discretize the phases of the characters in Γ :

$$u \longmapsto (e_p(\xi u))_{\xi \in \Gamma}.$$

This partitions $\mathbb{F}_p$ into boundedly many cells.

If u, v lie in the same cell, then

$$u - v \in B.$$

Therefore all edges inside a cell come from

$$A \cap B.$$

Since $|A \cap B| = O_{\mathcal{L}}(1)$, each cell has bounded maximum degree, hence bounded chromatic number.

Negative direction: statement

Assume no zero-sum subcollection has size at least 3.

We need to construct, for every t ,

$$A \subseteq \mathbb{F}_p$$

such that

$$|A| \geq \varepsilon p, \quad A \text{ is } \mathcal{L}\text{-solution-free}, \quad \chi(\text{Cay}(\mathbb{F}_p, A)) \geq t.$$

If there is no zero-sum subcollection at all, the Ramsey–Turan type theorem already gives such examples.

So the new case is the cancelling-pair case. After relabeling, suppose $c_1 + c_2 = 0$.

A Hamming-ball theorem with two consequences

Fix a prime q and let

$$S = \{\mathbf{x} \in \mathbb{Z}_q^n : \text{dist}(\mathbf{x}, \mathbf{1}) \leq q\sqrt{n}\}.$$

Theorem

$$\chi(\text{Cay}(\mathbb{Z}_q^n, S)) \geq \frac{\sqrt{n}}{q^3}, \quad \alpha(\text{Cay}(\mathbb{Z}_q^n, S)) = \Omega_q(q^n) \quad (n \rightarrow \infty).$$

- **Resolving Griesmer's question.** The first bound gives a quantitative yes: the high-chromatic Hamming-ball phenomenon persists for every fixed odd prime q .
- **Dynamical application.** With additional dynamical machinery, in every countably infinite abelian group, there is a topologically recurrent set that is not measurably recurrent.

Warm-up: classical Kneser philosophy

For $q = 2$, high chromaticity is related to Kneser graphs.

A k -set $A \subseteq [n]$ is encoded by its indicator vector $\mathbf{1}_A \in \mathbb{Z}_2^n$.

If $A \cap B = \emptyset$, then

$$\mathbf{1}_A - \mathbf{1}_B = \mathbf{1}_A + \mathbf{1}_B$$

is close to the all-ones vector when $2k$ is close to n .

Let $E = \text{Hamm}(\mathbf{1}, n - 2k)$, then the classical Kneser graph embeds into the Cayley graph $\text{Cay}(\mathbb{Z}_2^n, E)$.

Why characteristic 2 is not enough

For $x - y + z = 0$, a small Hamming ball around $\mathbf{1}$ is solution-free and contains the classical Kneser obstruction.

Now consider

$$\mathcal{L} : \quad x_1 - x_2 + x_3 + x_4 = 0.$$

Its coefficients contain a cancelling pair, but no zero-sum subcollection of size at least 3.

Over $\mathbb{Z}_2^n$, the equation becomes $x_1 + x_2 + x_3 + x_4 = 0$, and the radius-two Hamming ball around $\mathbf{1}$ already contains the distinct solution

$$\mathbf{1}, \quad \mathbf{1} + \mathbf{e}_1, \quad \mathbf{1} + \mathbf{e}_2, \quad \mathbf{1} + \mathbf{e}_1 + \mathbf{e}_2.$$

Thus the binary construction cannot handle the full cancelling-pair regime; an odd-characteristic obstruction is needed.

Why the familiar Kneser objects do not suffice

The classical embedding is binary

For disjoint A, B ,

$$\mathbf{1}_A - \mathbf{1}_B \in \{0, 1, -1\}^n.$$

For odd q , the two signs remain distinct, and disjointness no longer gives the required Hamming geometry.

The q -fold extension is a hypergraph

The standard q -fold Kneser generalization is a q -uniform hypergraph, whereas adjacency in a Cayley graph is pairwise.

A hypergraph chromatic lower bound therefore does not by itself transfer to $\text{Cay}(\mathbb{Z}_q^n, S)$.

The missing bridge

We need a genuinely q -ary *ordinary graph* whose every edge maps to an edge of $\text{Cay}(\mathbb{Z}_q^n, S)$.

A generalized Kneser graph

For $n \geq (m+1)k$, let $\text{KN}(n, k, m)$ be the graph whose vertices are ordered tuples

$$\mathcal{A} = (A_1, \dots, A_m), \quad |A_i| = k, \quad A_i \cap A_j = \emptyset \quad (i \neq j).$$

Two vertices $\mathcal{A} = (A_1, \dots, A_m)$ and $\mathcal{B} = (B_1, \dots, B_m)$ are adjacent if, after possibly interchanging $\mathcal{A}$ and $\mathcal{B}$,

$$\left(\bigcup_{\ell=1}^i A_\ell \right) \cap \left(\bigcup_{\ell=i}^m B_\ell \right) = \emptyset \quad \text{for every } i \in [m].$$

When $m = q - 1$, the ordering records the nonzero symbols of $\mathbb{Z}_q$; the prefix–suffix rule is tailored to the Cayley embedding.

Embedding into the Hamming-ball Cayley graph

Embed a vertex of $\text{KN}(n, k, q - 1)$ into $\mathbb{Z}_q^n$ by

$$(A_1, \dots, A_{q-1}) \mapsto \mathbf{1}_{A_1} + 2\mathbf{1}_{A_2} + \dots + (q - 1)\mathbf{1}_{A_{q-1}}.$$

If two vertices are adjacent in the generalized Kneser graph, their images have difference close to $\mathbf{1}$.

Hence the generalized Kneser graph embeds into

$$\text{Cay}(\mathbb{Z}_q^n, S).$$

So a chromatic lower bound for the generalized Kneser graph transfers to the Cayley graph.

Topological lower bound

Suppose the generalized Kneser graph has a proper coloring with few colors.

We place n labeled points on a complex sphere

$$\mathbb{S}^{2r-1} \subseteq \mathbb{C}^r.$$

For each point $x \in \mathbb{S}^{2r-1}$, the orbit

$$x, \zeta x, \zeta^2 x, \dots, \zeta^{q-1} x$$

cuts the sphere into q cyclic open sectors

$$V_0(x), V_1(x), V_2(x), \dots, V_{q-1}(x).$$

Topological lower bound

If each sector contains enough labels, then there exists a vertex

$$(A_1, \dots, A_{q-1})$$

such that

$$A_1 \subseteq V_1(x), A_2 \subseteq V_2(x), \dots, A_{q-1} \subseteq V_{q-1}(x).$$

Thus each color class gives an open set U_j on the sphere:

$$x \in U_j \iff x \text{ realizes a vertex of color } j.$$

Topological lower bound

Lemma

Suppose $q(k + 2qr) \leq n$. One may find n labeled points on the sphere $\mathbb{S}^{2r-1}$ such that for every $x \in \mathbb{S}^{2r-1}$ and $i \in \{0, 1, \dots, q-1\}$, the sector $V_i(x)$ contains at least k labeled points.

By taking such points, the sets $U_1, \dots, U_t$ cover the sphere.

Lemma

Let $U_1, \dots, U_t$ be open sets that cover $\mathbb{S}^{2r-1}$. If $(t-1)(q-1) < 2r$, then one of the sets $U_1, \dots, U_t$ contains a full $\mathbb{Z}_q$ -orbit $\{x, \zeta x, \dots, \zeta^{q-1}x\}$.

Topological lower bound

By the Borsuk–Ulam type theorem, if t is too small, then some U_j contains a full $\mathbb{Z}_q$ -orbit:

$$x, \zeta x, \dots, \zeta^{q-1}x \in U_j.$$

Take witnesses for

$$x \in U_j \quad \text{and} \quad \zeta x \in U_j.$$

Because the sectors are cyclically shifted, these two witnesses give adjacent vertices of the generalized Kneser graph.

So we find a monochromatic edge, contradicting proper coloring.

$\chi(\text{KN}(n, k, q - 1))$ is large.

What the obstruction gives us

Combining the embedding with the topological lower bound gives

$$\chi(\text{Cay}(\mathbb{Z}_q^n, S)) \geq \frac{\sqrt{n}}{q^3},$$

where

$$S = \{\mathbf{x} \in \mathbb{Z}_q^n : \text{dist}(\mathbf{x}, \mathbf{1}) \leq q\sqrt{n}\}.$$

This is the high-chromatic object used in the construction.

Construction for the negative direction

Assume the only zero-sum obstruction is a cancelling pair:

$$c_1 + c_2 = 0, \quad \text{no zero-sum subcollection has size } \geq 3.$$

We first work in a product group

$$\mathbb{Z}_m \cong \prod_{i=1}^n \mathbb{Z}_{p_i}.$$

Construct $E_0 \subseteq \mathbb{Z}_m$ such that

$$\chi(\text{Cay}(\mathbb{Z}_m, E_0)) \text{ is large,} \quad E_0 \text{ is } \mathcal{L}\text{-solution-free.}$$

The idea is:

Hamming-ball obstruction + asymmetric norm separation.

Making the construction dense

The high-chromatic set E_0 may be sparse, so we add a dense extension set $F_0 \subseteq \mathbb{Z}_m$ with

$$|F_0| = \Theta(m).$$

It is chosen so that

$$(-c_1 F_0 - c_2 F_0) \cap (c_3 E_0 + \cdots + c_k E_0) = \emptyset.$$

This blocks the only dangerous cancellation between the dense part and the high-chromatic part.

Finally, lift to a much larger prime field $\mathbb{F}_p$, and set

$$A = E \sqcup F.$$

Then

$$|A| \geq \varepsilon p, \quad A \text{ is } \mathcal{L}\text{-solution-free}, \quad \chi(\text{Cay}(\mathbb{F}_p, A)) \geq t.$$

Thank you

Thank you!